Benchmarking Deep Reinforcement Learning Algorithms for Portfolio Optimization

Ossama Adel Moursy
ID: 20595449
Dept. of Computing
Queen's University
ossama.m@queensu.ca

Abdelrahman Ibrahim Kamal Eldin
ID: 20596377
Queen's University
abdelrahman.ibrahim@queensu.ca

Aya Said Abdallah Noah
ID: 20596338
Dept. of Computing
Queen's University
25xgvp@queensu.ca

Esraa Mohammed Elsayed
ID: 20596301
Dept. of Computing
Queen's University
esraa.farag@queensu.ca

CISC 856 - Reinforcement Learning (Group 7), School of Computing, Queen's University, Kingston, Ontario, Canada
Instructor: Dr. Sidney Givigi

***Abstract*—Portfolio optimization deciding how to split capital across assets to maximize return for an acceptable level of risk has classically been solved with the Markowitz mean-variance model, whose assumptions of stationary and predictable markets are routinely violated in practice. Deep Reinforcement Learning (DRL) offers a model-free alternative that learns an allocation policy directly from market experience. Rather than proposing a single agent, this work conducts a controlled benchmark of four widely used DRL algorithms PPO, A2C, SAC, and TD3 on a daily allocation task over seven assets plus cash, holding the environment, reward, feature set, and training budget identical so that any performance gap is attributable to the algorithm alone. The agents are compared against three non-learning baselines (equal-weight 1/N, the long-only Markowitz tangency portfolio, and a passive buy-and-hold of the market). Across a grid of 72 training runs (4 algorithms × 3 risk-aversion levels × 2 look-back windows × 3 seeds), the on-policy methods clearly dominate the off-policy ones: PPO is the strongest agent (test Sharpe 0.87 ± 0.10), followed by A2C (0.75), while SAC (0.47) and TD3 (0.42) struggle to learn within the budget. None of the agents surpasses the classical baselines on a risk-adjusted basis (1/N Sharpe 0.98, Markowitz 0.95) over the full 2022-2024 test period, and in a concrete $10,000 one-year bear-market simulation the Markowitz portfolio ends highest. We analyse why on-policy learning generalizes better here, how risk aversion and look-back length affect each algorithm, and what the learned allocations actually do, providing a reproducible reference point for DRL portfolio studies.**

Index Terms*—*Reinforcement learning, portfolio optimization, PPO, A2C, SAC, TD3, benchmarking, algorithmic trading, Sharpe ratio.

I. Introduction

A central problem in quantitative finance is portfolio optimization: given a universe of assets, how should an investor allocate capital to maximize return while controlling risk? The foundational answer is Markowitz's mean-variance optimization (MVO) [1], which selects the weights offering the best expected return for a chosen variance. Despite its elegance, MVO rests on assumptions stationary return distributions, reliably estimable means and covariances that real markets violate prices move unpredictably, volatility clusters and spikes, and the relationships that held last year may not hold this year. Estimation error in the inputs, especially expected returns, is well known to make MVO portfolios unstable and concentrated.

These weaknesses motivate a model-free approach. Deep Reinforcement Learning (DRL) frames allocation as a sequential decision problem: an agent observes the market and its own holdings, chooses a new allocation, receives a reward tied to profit and risk, and improves its policy from experience [2], [3]. Crucially, the agent need not assume a return model; it learns one implicitly from data.

A practical difficulty for anyone adopting DRL for this task is which algorithm to use. The literature reports results for many different agents, but usually one algorithm per study, trained under bespoke environments, reward functions, and compute budgets, which makes cross-paper comparison unreliable. This work takes a benchmarking stance: we fix a single trading environment, reward, observation set, and step budget, and run four popular DRL algorithms from the same library through an identical experimental grid. Any difference we observe is therefore due to the learning algorithm rather than the surrounding setup [3].

Concretely, our contributions are:

- A common Gymnasium environment for long-only daily allocation over seven assets plus cash, with a causal feature set, a softmax action that includes an explicit cash position, and a risk-adjusted reward with a turnover penalty (Section III).
- A controlled comparison of PPO, A2C, SAC, and TD3 over 72 runs against three non-learning baselines, reported with genuine across-seed error bars and seven risk/return metrics (Sections IV-V).
- An analysis of why the on-policy methods outperform the off-policy ones in this regime, how risk aversion λ and look-back length affect each algorithm, what the best agent's learned allocation looks like, and an honest account of how the agents compare to classical baselines in both a full-period and a one-year investment simulation.

II. Related Work

Classical allocation. Mean-variance optimization [1] remains the reference framework, and the maximum-Sharpe (tangency) portfolio [4] is its canonical risk-adjusted solution. Its sensitivity to input estimation error is the practical limitation that learning-based methods aim to sidestep.

Reinforcement learning for trading. Direct reinforcement was applied to trading by Moody and Saffell [5], who optimized a differentiable Sharpe-like objective without a separate forecasting step. Deng et al. [6] combined deep feature learning with recurrent reinforcement for single-asset trading. For portfolio management, Jiang et al. [2] introduced an influential deterministic-policy framework over multiple assets, and the FinRL library [3] standardized environments and pipelines for automated stock trading, explicitly arguing for controlled comparison of algorithms the same motivation we adopt here.

The algorithms under test. We benchmark two on-policy and two off-policy actor-critic methods. Proximal Policy Optimization (PPO) [7] stabilizes policy-gradient updates with a clipped surrogate objective; Advantage Actor-Critic (A2C) [8] is the synchronous variant of A3C. Both learn on policy from freshly collected trajectories. Soft Actor-Critic (SAC) [9] is an off-policy maximum-entropy method that reuses past experience through a replay buffer and automatically tunes its exploration temperature, while Twin Delayed DDPG (TD3) [10] is a deterministic off-policy method that uses clipped double-Q learning and delayed policy updates to curb value overestimation. All four are taken from Stable-Baselines3 [11] with a multilayer-perceptron policy, and the environment is built on Gymnasium [12]. The gap our study addresses is the scarcity of head-to-head evaluations of these algorithms under truly identical financial conditions.

III. Methodology

We model daily portfolio allocation as a Markov Decision Process and train each agent to choose, once per trading day, a vector of portfolio weights. The acting reward timing is causal: the agent acts at day t using only information available at t, and the return is realized at t+1.

A. State (Observation) Space

Let N = 7 be the number of risky assets. For each asset we compute four summary features the 1-, 5-, and 21-day log-returns and the 21-day rolling volatility each standardized with a causal trailing-window z-score that uses only past data:

$$z(x_t) = \frac{x_t - \mu_{t-W:t}}{\sigma_{t-W:t} + \epsilon}, W = 252,\ \epsilon = 10^{-8} \tag{1}$$

Stacking the F = 4 features across the N assets and concatenating the current weight vector over the risky assets and cash gives the observation

$$s_t = [z(features_t), w_t] \in \mathbb{R}^{FN+(N+1)} = \mathbb{R}^{36} \tag{2}$$

Because all normalization statistics are trailing, no information from the test period leaks into training. The environment also supports a high-dimensional “full” observation built from raw price windows; we discuss why the compact form is used in Section III-D.

B. Action Space

At each step the policy outputs an unconstrained vector $a_t \in \mathbb{R}^{N+1}$, whose last entry is a cash position. A softmax maps it to a long-only, fully invested portfolio:

$$w_t = softmax(a_t),\ w_{t,i} \geq 0,\ \textstyle\sum_i w_{t,i} = 1 \tag{3}$$

The softmax simultaneously enforces the no-short-selling and budget constraints regardless of the raw scale produced by a given algorithm, so PPO, A2C, SAC, and TD3 all yield valid portfolios. Modelling cash as a real allocatable asset (earning the risk-free rate r_f, set to 0) lets the agent actively de-risk by moving into cash rather than treating cash as a dead input.

C. Reward Function

The per-step reward is a risk-adjusted return net of trading cost:

$$R_t = \left(\textstyle\sum_i w_{t,i}\, \rho_{t+1,i} + w_{cash} r_f \right) - \lambda \sigma_t^p - \kappa \parallel w_t^r - w_{t-1}^r \parallel_1 \tag{4}$$

where ρ_{t+1} are next-day simple returns, σ_t^p is the rolling standard deviation of recent portfolio returns (a 20-day window), $\lambda \in \{0, 0.5, 1\}$ is the risk-aversion coefficient (λ = 0 is a pure profit seeker, λ = 1 is strongly risk-averse), and κ = 0.1% is a transaction cost charged on risky-asset turnover. The cost term discourages pointless churn and mimics real-market friction.

D. Environment

The MDP above is implemented as a Gymnasium environment in which one agent step equals one trading day. Each episode runs across a fixed date range (the training or the test window), starting fully in cash. The same action and reward are shared by both observation modes. We deliberately use the compact 36-dimensional observation rather than the full windowed form: full price windows inflate the observation to roughly 640 dimensions, which makes the off-policy methods (SAC, TD3) prohibitively slow to train on CPU. The compact features bring all four algorithms into a feasible budget so that the benchmark is complete, a deliberate scope choice consistent with avoiding problems that take too long to train.

E. Baselines and Evaluation Metrics

We compare the agents against three non-learning strategies, all paying the same 0.1% cost and using only past data: (i) equal-weight (1/N), which rebalances to uniform weights daily; (ii) Markowitz (max-Sharpe), the long-only tangency portfolio re-estimated from a trailing 252-day window and rebalanced every 21 trading days,

$$w^* = \arg\max_w \frac{w^\top \mu}{\sqrt{w^\top \Sigma w}} \quad s.t. \textstyle\sum_i w_i = 1, w_i \geq 0 \tag{5}$$

with μ, Σ annualized; and (iii) a passive buy-and-hold of the market index (SPY).

Performance is summarized with annualized metrics (252 trading days): total return, compound annual growth rate (CAGR), volatility, the Sharpe and Sortino ratios (the latter penalizing only downside deviation), maximum drawdown (MDD), and the Calmar ratio (CAGR over |MDD|):

$$\text{Sharpe} = \frac{\bar{r}}{\bar{\sigma}_r}\sqrt{252}\,, \qquad \text{Calmar} = \frac{\text{CAGR}}{|\text{MDD}|} \tag{6}$$

TABLE I
EXPERIMENTAL CONFIGURATION (SHARED ACROSS ALL RUNS)

Component	**Setting**
Algorithms	PPO, A2C (on-policy); SAC, TD3 (off-policy) [11]
Policy network	MLP (MlpPolicy), SB3 defaults
Timesteps / run	50,000 (CPU)
Off-policy extras	buffer 50,000; learning starts 1,000
Risk aversion λ	{0.0, 0.5, 1.0}
Look-back L	{30, 60} days
Seeds	{0, 1, 2}
Transaction cost κ	0.1% of risky turnover
Observation dim.	36 (compact, causal z-score)
Action dim.	8 (7 assets + cash, softmax)
Total runs	72

IV. EXPERIMENTAL SETUP

Data. Daily adjusted prices for seven assets four technology stocks (AAPL, MSFT, NVDA, AMZN), gold (GLD), oil (USO), and the S&P 500 index (SPY) were downloaded from Yahoo Finance for 2015-2024. After computing the feature series, 2,494 trading days remain (2015-02-03 to 2024-12-30). We use a single chronological split at 2022-01-01, giving 1,742 training days and 752 out-of-sample test days; all test results are strictly out of sample.

Agents and grid. Each algorithm uses the Stable-Baselines3 [11] MlpPolicy with default hyperparameters and trains for 50,000 timesteps on CPU. SAC and TD3 additionally use a replay buffer of 50,000 and 1,000 warm-up steps before learning. We sweep three risk-aversion levels λ ∈ {0, 0.5, 1} and two look-back windows L ∈ {30, 60}, and repeat every configuration with three random seeds, for a total of 4 × 3 × 2 × 3 = 72 runs (Table I). Three seeds let the reported error bars reflect genuine run-to-run variance rather than noise across unrelated settings. Every run log to TensorBoard and a Monitor CSV, and finished runs are checkpointed, so the grid is resumable.

Sanity check. Before the full grid we trained PPO briefly (20,000 steps) to confirm that the whole pipeline, from data loading to the environment to the agent, is wired correctly; this short run reached a test Sharpe of 0.86 against 0.98 for 1/N, confirming the pipeline functions while not yet being competitive.

V. RESULTS AND DISCUSSION

A. The Asset Universe

Figure 1 shows the normalized price paths and the daily-return correlation. NVDA dominates the period (growing several hundred-fold), the technology names are strongly correlated with each other and with SPY, while GLD is nearly uncorrelated with everything, a genuine diversifier. Figure 2 places each asset in risk/return space and tracks rolling volatility: NVDA offers the highest return at the highest risk, GLD sits at low risk/low return, USO carries high volatility for almost no return over the decade, and every asset's volatility spikes in early 2020. These properties matter for interpretation: a sensible allocator should lean on the tech-gold mix and avoid over-weighting USO, and any agent must cope with a non-stationary, regime-switching volatility environment.

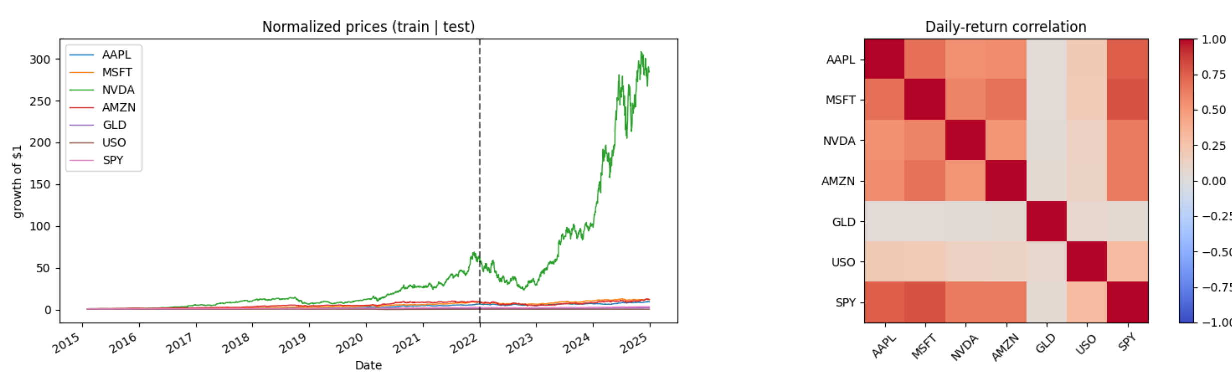

Fig. 1. *Dataset.* Left: normalized price paths (growth of $1) with the train/test split (dashed) at 2022-01-01; NVDA dominates the universe. Right: daily-return correlation matrix technology names co-move strongly while GLD is nearly uncorrelated.

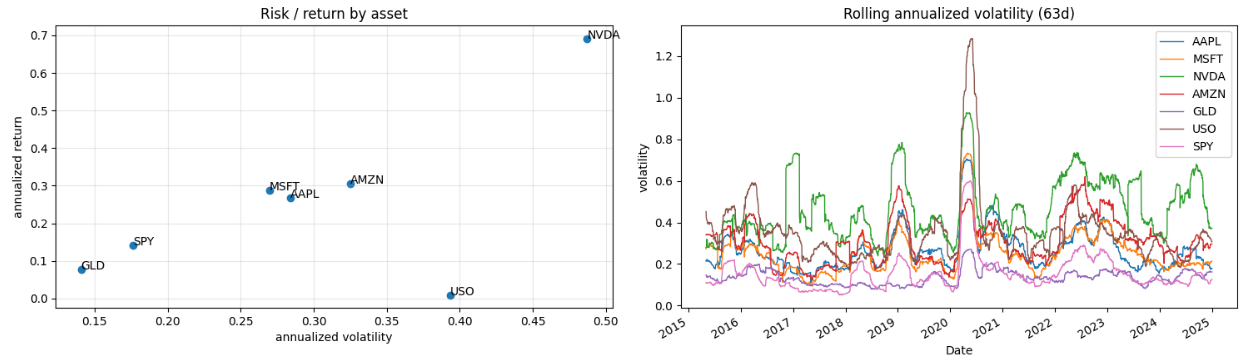

Fig. 2. *Asset characteristics.* Left: annualized risk/return by asset NVDA is high-risk/high-return, GLD low-risk, USO high-risk/near-zero-return. Right: 63-day rolling annualized volatility, showing the early-2020 spike and persistent regime changes.

B. Algorithm Comparison

Table II reports the best configuration of each algorithm (selected by mean test Sharpe over its three seeds) alongside the baselines, and Figure 3 visualizes the Sharpe ranking with a cross-seed error bar. The ordering among the agents is unambiguous: PPO is best (Sharpe 0.87 ± 0.10, 55.7% total return over the test window), followed by A2C (0.75), then SAC (0.47) and TD3 (0.42). The best single run overall is PPO with $\lambda = 0$, $L = 60$ at a Sharpe of 0.96 (Sortino 1.46, total return 64.7%). Off-policy methods not only score lower but also exhibit much wider seed-to-seed variance (SAC std 0.23, TD3 std 0.14), indicating unstable learning.

Comparing to the non-learning baselines is where honesty matters. Over the full 2022-2024 test period, neither baseline is beaten on a risk-adjusted basis: equal-weight (1/N) attains a Sharpe of 0.98 and the Markowitz tangency portfolio 0.95, both above PPO's 0.87, and Markowitz also delivers the highest total return (83.9%). The Sortino picture is similar (1.44 for both baselines versus 1.46 for the single best PPO run). The benchmark's value, however, is precisely this controlled verdict: among the learning methods PPO is clearly preferable, and a simple 1/N rule remains a remarkably strong, hard-to-beat baseline, a recurring finding in portfolio research.

TABLE II
OUT-OF-SAMPLE PERFORMANCE OVER THE FULL TEST PERIOD (2022-2024). AGENTS ARE THE BEST CONFIGURATION PER ALGORITHM (MEAN OVER 3 SEEDS); SHARPE SHOWN WITH A CROSS-SEED STD. BEST VALUE PER COLUMN IN BOLD.

Strategy	Config	Total Ret.		CAGR	Volatility	Sharpe	Max DD	Calmar
Non-learning baselines								
Equal weight (1/N)	-	74.8%		20.6%	21.4%	**0.98**	- 28.2%	0.73
Markowitz (max-Sharpe)	252d, monthly	**83.9%**		**22.7%**	24.8%	0.95	- 29.4%	**0.77**
DRL agents (best config / algorithm)								
PPO	λ=0.0, L=60	55.7%		16.0%	**19.1%**	0.87 ± 0.10	**- 27.2%**	0.59
A2C	λ=0.0, L=60	47.0%		13.8%	19.8%	0.75 ± 0.06	- 28.7%	0.49
SAC	λ=1.0, L=60	24.8%		7.6%	19.3%	0.47 ± 0.23	- 29.3%	0.27
TD3	λ=1.0, L=30	23.0%		7.1%	21.8%	0.42 ± 0.14	- 28.5%	0.25

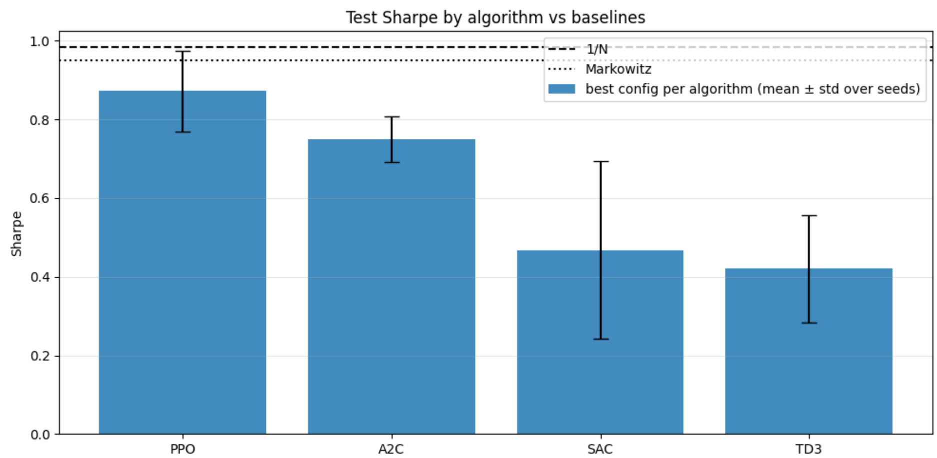

Fig. 3. Test Sharpe by algorithm (best configuration each, mean ± std over 3 seeds) against the 1/N and Markowitz baselines (horizontal lines). PPO leads the learning methods, but both classical baselines remain above all agents.

C. Sensitivity to Risk Aversion and Look-back

Figure 4 reports mean test Sharpe for every (λ, L) pair, one heat-map per algorithm. Three patterns stand out. First, PPO is uniformly strong (0.72-0.87) and the most robust to its settings: changing λ barely moves its Sharpe, which is desirable because it means the practitioner does not need to tune risk aversion delicately. Second, a longer look-back (L = 60) helps the on-policy methods PPO and A2C both peak at L = 60 suggesting they benefit from a wider view of recent dynamics. Third, the off-policy methods are erratic: SAC hovers around 0.4 with no clear trend, and TD3 ranges from a modest 0.42 down to -0.14 at λ = 1, L = 60, where the combination of strong risk penalty and longer window appears to destabilize its value estimates entirely. In short, the best agent (PPO) is also the easiest to configure, while the worst (TD3) is the most configuration sensitive.

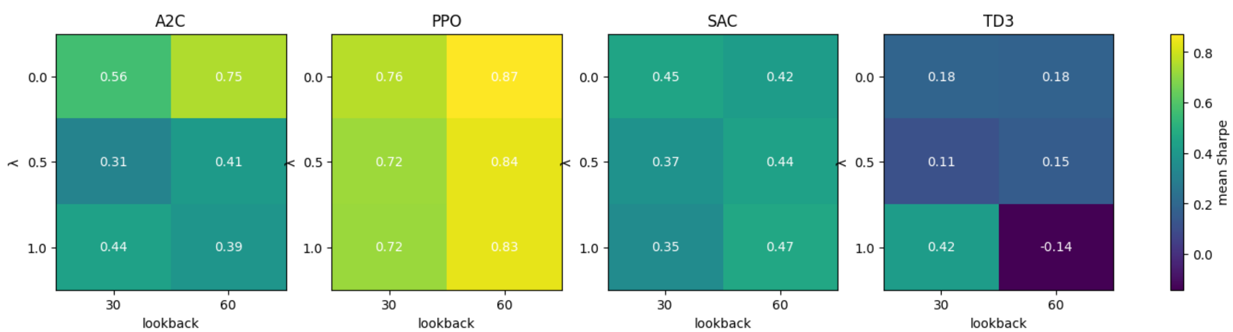

Fig. 4. Mean test Sharpe (over 3 seeds) for each risk-aversion λ (rows) and look-back L (columns), one heat-map per algorithm. PPO is consistently high and robust; TD3 is weak and even negative at λ = 1, L = 60.

D. Learning Dynamics: On-policy vs. Off-policy

The training curves in Figure 5 explains the ranking. PPO and A2C converge almost immediately to a small positive episode reward and stay there, whereas SAC and TD3 begin deeply negative (around -15) and climb only slowly TD3 toward -8.5 and SAC barely improving within the 50,000-step budget. We attribute this to the interaction between off-policy learning and the financial signal. Daily returns are dominated by noise, so the next-day reward is only weakly predictable; the off-policy critics in SAC and TD3 must bootstrap value estimates from a replay buffer of such noisy, non-stationary transitions, which is exactly the setting in which value overestimation and instability arise. On-policy methods, by contrast, optimize the policy directly from fresh trajectories and never bootstrap a long-horizon value across regime changes, making them better suited to this short-budget, high-noise problem. This is consistent with the well-known higher sample efficiency but greater tuning fragility of off-policy actor-critics, and it is plausible that SAC/TD3 would close the gap with substantially more steps or task-specific tuning, an avenue we leave to future work.

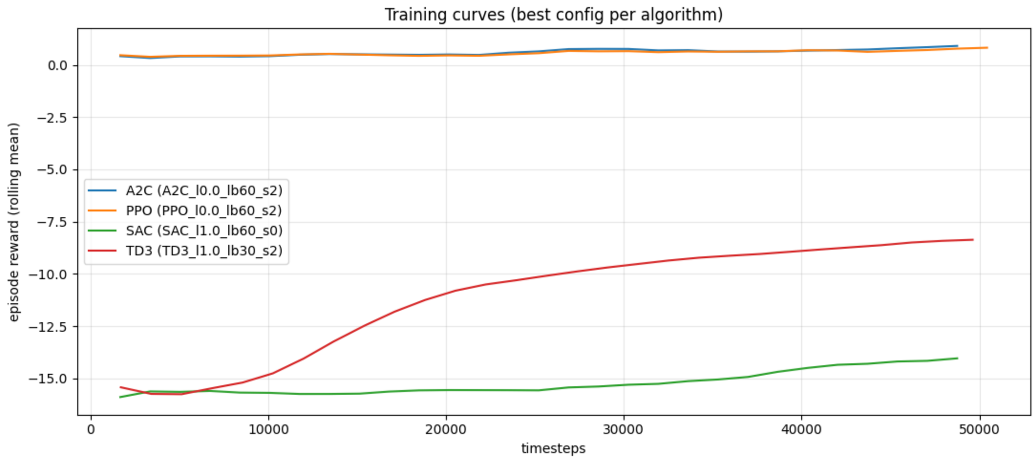

Fig. 5. Smoothed episode reward during training for the median-seed run of each algorithm's best configuration. The on-policy methods (PPO, A2C) are stable from the start; the off-policy methods (SAC, TD3) struggle to learn within 50,000 steps.

E. What the Best Agent Does, and Out-of-sample Equity

Figure 6 plots out-of-sample cumulative growth for each algorithm's best configuration against the baselines. All strategies fall together through the 2022 drawdown and recover from mid-2023; the baselines pull ahead in the 2024 rally, with PPO the leading learning method and SAC/TD3 trailing. Figure 7 dissects the best PPO agent's behaviour. Rather than concentrating, it holds a broadly diversified mix and maintains an active cash sleeve (grey) that expands during turbulent stretches, and its drawdown reaches roughly -26% during the 2022 early 2023 trough before recovering behaviour qualitatively consistent with a risk-aware allocator, even though it does not out-earn the simpler rules. The agent also keeps day-to-day turnover moderate, which limits the 0.1% cost drag.

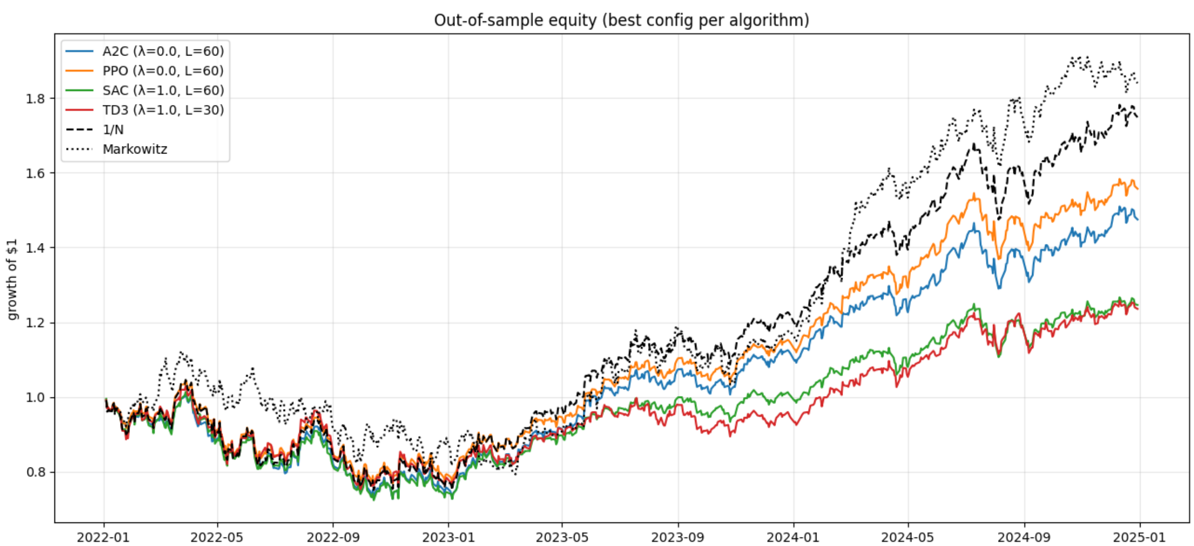

Fig. 6. Out-of-sample equity (growth of $1) for the best configuration of each algorithm (averaged over its seeds) versus the 1/N and Markowitz baselines over the full 2022-2024 test period.

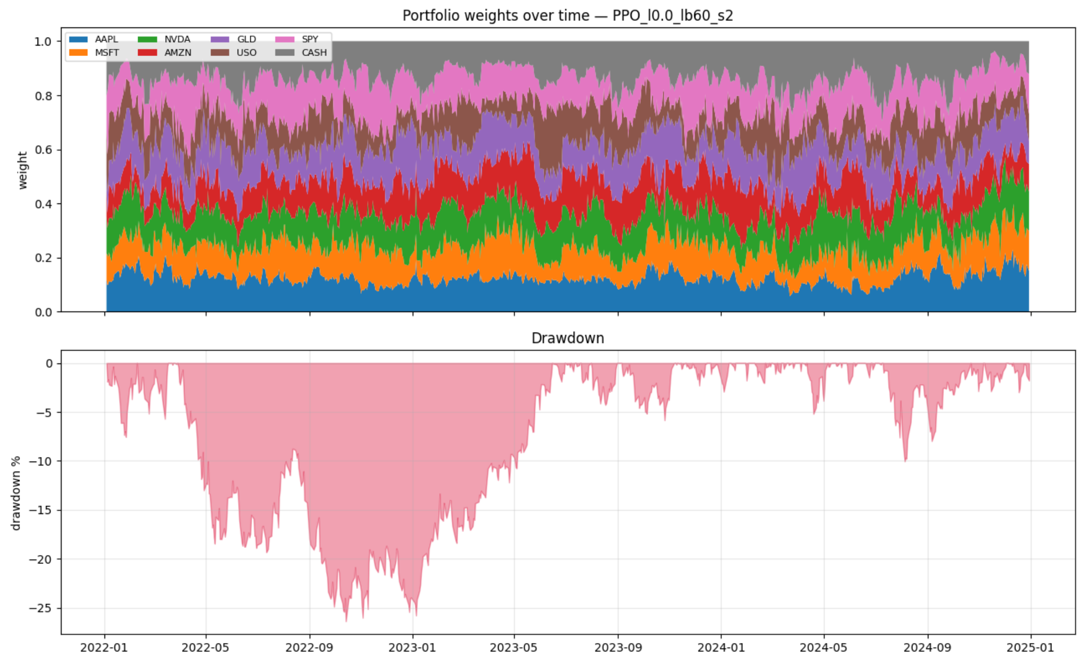

Fig. 7. Behaviour of the best PPO agent (median seed). Top: portfolio weights over time including the cash sleeve (grey), which grows during turbulent periods. Bottom: drawdown, reaching about −26% in the 2022 early 2023 trough.

F. Investment Simulation: $10,000 over One Year

To make the comparison tangible, Table III and Figure 8 track a $10,000 investment over the first 252 test days calendar year 2022, a pronounced bear market. Ranked by final value, the Markowitz portfolio ends highest ($8,167,

-18.3%) and has the best Sharpe (−0.51), the buy-and-hold of SPY is a hair behind ($8,164, −18.4%), the RL (PPO) agent finishes at $7,825 (-21.8%), and equal-weight is last ($7,638, -23.6%). Because the returns are negative here, a smaller (less negative) number is the better outcome, a smaller loss, and the maximum-drawdown column reports the worst peak-to-trough dip during the year rather than the final loss.

The year 2022 was a broad market downturn in which essentially every asset in the universe fell; because all strategies are constrained to be long-only and fully invested, the capital loss here reflects the market regime itself rather than a failure of any method.

A ranking over a single down-year also differs from the risk-adjusted, full-period result: the same PPO agent that finishes behind here returns +55.7% over the complete 2022-2024 window once the recovery is included and leads the other three RL algorithms throughout. In that sense the model succeeds at its objective, learning a diversified, cost- and risk-aware allocation policy even though it does not beat the simplest baselines over this short, falling slice.

Finally, the shaded seed band in Figure 8 shows that the outcome moves noticeably with the random seed alone, a reminder that single-window results should be read with care.

TABLE III
$10,000 INVESTED OVER ONE YEAR (252 TEST DAYS, 2022 BEAR MARKET). RANKED BY FINAL VALUE.

Strategy	**Final $**	**ROI**	**Max DD**	**Sharpe**
Markowitz (max-Sharpe)	**8,167**	**-18.3%**	-29.0%	**-0.51**
Buy & hold SPY	8,164	-18.4%	**-24.5%**	-0.72
RL model (PPO)	7,825	-21.8%	-26.7%	-0.86
Equal weight (1/N)	7,638	-23.6%	-28.2%	-0.80

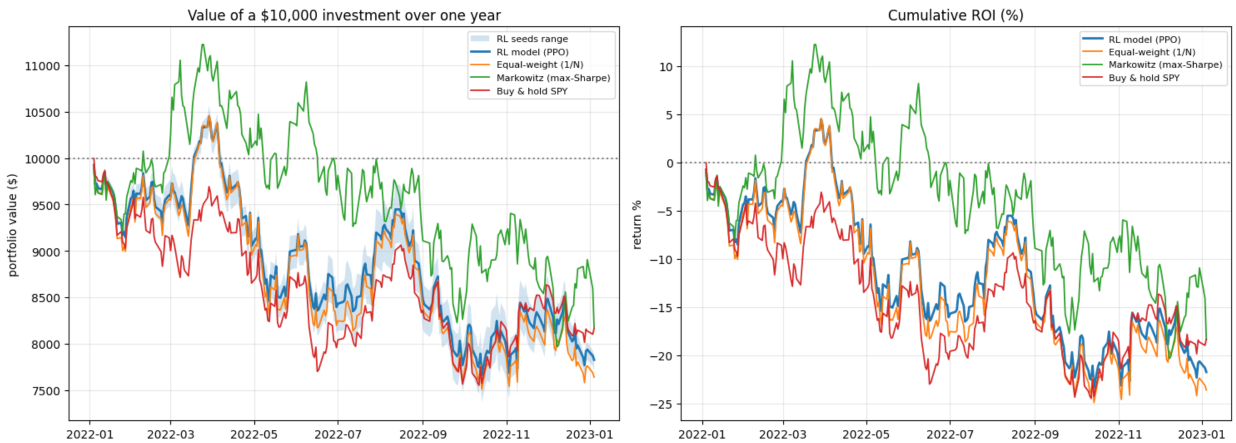

Fig. 8. A $10,000 investment over one year (2022). Left: dollar value, with the shaded band giving the spread of the PPO agent across seeds. Right: cumulative ROI. All strategies lose ground in this bear market; Markowitz ends highest.

G. Discussion and Limitations

The headline message is that, under identical conditions, PPO is the most reliable DRL algorithm for this task and on-policy learning clearly beats off-policy learning within a modest budget, yet none of the agents dethrones a well-constructed classical baseline on a risk-adjusted basis. Several factors plausibly cap the agents' edge. The compact observation, chosen for tractability, discards cross-sectional and fine temporal structure that a richer encoder might exploit; the long-only, fully-invested action space removes shorting and leverage, narrowing the room to out-perform; the daily reward is extremely noisy; and a 50,000-step budget is short for actor-critic methods, especially off-policy ones. The evaluation also rests on a single chronological split and one asset universe, so the numbers describe this regime rather than a universal verdict. We made these scope choices deliberately so that the full four-algorithm grid could be completed and compared fairly; the result is a clean, reproducible benchmark rather than a state-of-the-art trading system.

VI. Conclusion

We presented a controlled benchmark of four deep reinforcement learning algorithms: PPO, A2C, SAC, and TD3 for long-only daily portfolio allocation over seven assets plus cash, holding the environment, reward, features, and budget fixed so that performance differences are attributable to the algorithm alone. Across 72 runs, PPO was the strongest agent (test Sharpe 0.87 ± 0.10) and on-policy methods decisively outperformed off-policy ones, which struggled to learn from the noisy financial signal within the budget. PPO was also the most robust to its risk-aversion and look-back settings. Against non-learning baselines, however, the classical equal-weight and Markowitz portfolios remained ahead on a risk-adjusted basis over the full test period, and in a one-year bear-market simulation the Markowitz portfolio ended highest, an honest reflection of how hard these baselines are to beat.

Future work follows directly from our scope choices: train the off-policy methods for far longer (or with task-specific tuning) to test whether they close the gap; replace the compact features with a sequence encoder (LSTM or transformer) over full price windows; broaden the asset universe and evaluate across multiple market regimes with walk-forward splits; incorporate transaction-cost-aware and drawdown-aware objectives directly; and explore agent ensembles. The environment, configurations, and per-run results released with this study provide a reproducible reference point for such extensions.

Acknowledgement

We thank Dr. Sidney Givigi for his guidance throughout CISC 856, and the course teaching assistant for helpful feedback on framing the experiments. All experiments used the open-source Gymnasium [12] and Stable-Baselines3 [11] libraries.

References

[1] H. Markowitz, "Portfolio selection," The Journal of Finance, vol. 7, no. 1, pp. 77-91, 1952.

[2] Z. Jiang, D. Xu, and J. Liang, "A deep reinforcement learning framework for the financial portfolio management problem," arXiv:1706.10059, 2017.

[3] X.-Y. Liu et al., "FinRL: A deep reinforcement learning library for automated stock trading in quantitative finance," in NeurIPS Deep RL Workshop, 2020.

[4] W. F. Sharpe, "Mutual fund performance," The Journal of Business, vol. 39, no. 1, pp. 119-138, 1966.

[5] J. Moody and M. Saffell, "Learning to trade via direct reinforcement," IEEE Trans. Neural Networks, vol. 12, no. 4, pp. 875-889, 2001.

[6] Y. Deng, F. Bao, Y. Kong, Z. Ren, and Q. Dai, "Deep direct reinforcement learning for financial signal representation and trading," IEEE Trans. Neural Networks and Learning Systems, vol. 28, no. 3, pp. 653-664, 2017.

[7] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," arXiv:1707.06347, 2017.

[8] V. Mnih et al., "Asynchronous methods for deep reinforcement learning," in Proc. ICML, 2016, pp. 1928-1937.

[9] T. Haarnoja, A. Zhou, P. Abbeel, and S. Levine, "Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor," in Proc. ICML, 2018.

[10] S. Fujimoto, H. van Hoof, and D. Meger, "Addressing function approximation error in actor-critic methods," in Proc. ICML, 2018.

[11] A. Raffin, A. Hill, A. Gleave, A. Kanervisto, M. Ernestus, and N. Dormann, "Stable-Baselines3: Reliable reinforcement learning implementations," Journal of Machine Learning Research, vol. 22, no. 268, pp. 1-8, 2021.

[12] M. Towers et al., "Gymnasium," Zenodo, 2023. [Online]. Available: https://zenodo.org/record/8127025